

SETTING UP NETWORKING SIMULATION TOOLS AND INSTALLING WIRESHARK FOR NETWORK TRAFFIC ANALYSIS

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1. Introduction

Network traffic analysis is an essential skill in cybersecurity, network administration, and digital forensics. Organizations use packet analysis tools to monitor communications, detect malicious activities, troubleshoot connectivity issues, and understand how protocols operate within a network.

This project focuses on the installation and configuration of Wireshark, a widely used network protocol analyzer, and the use of packet capture techniques to inspect live network traffic. The project also demonstrates protocol identification and analysis using real network communications.

2. Project Overview

The purpose of this project was to establish a network monitoring environment capable of capturing and analyzing data packets traversing a network.

The project introduced the use of Wireshark to:

- Capture network packets.
- Analyze communication protocols.
- Observe network behavior.
- Perform basic troubleshooting.
- Develop practical cybersecurity monitoring skills.

3. Project Objectives

The objectives of this project were:

1. Install Wireshark successfully.
2. Configure packet capture drivers.
3. Identify active network interfaces.
4. Capture live network traffic.
5. Analyze common protocols.
6. Understand packet structure.
7. Gain practical experience with cybersecurity monitoring tools.

4. Scope of the Project

This project falls under:

Cybersecurity Monitoring and Network Traffic Analysis

The scope includes:

- Network packet capture.
- Protocol analysis.
- Security monitoring.
- Network troubleshooting.
- Traffic inspection.

The project does not involve offensive security activities or penetration testing.

5. Tools and Software Used

Tool	Purpose
Wireshark 4.6.6	Packet Capture and Analysis
Npcap	Packet Capture Driver
Windows Operating System	Host Environment
Network Adapter (Wi-Fi 2)	Traffic Source

6. Assets, Threats, and Vulnerabilities

Assets

The following assets were identified:

- Computer System
- Wi-Fi Network Adapter
- Network Traffic
- Internet Connectivity
- Packet Capture Files

Threats

Potential threats include:

- Packet sniffing
- Man-in-the-middle attacks
- Data interception
- Unauthorized monitoring
- Malware communications

Vulnerabilities

Potential vulnerabilities include:

- Unencrypted traffic
- Weak network configurations
- Lack of monitoring
- Misconfigured systems

7. Wireshark Installation and Configuration

Wireshark was downloaded from the official Wireshark website and installed successfully.

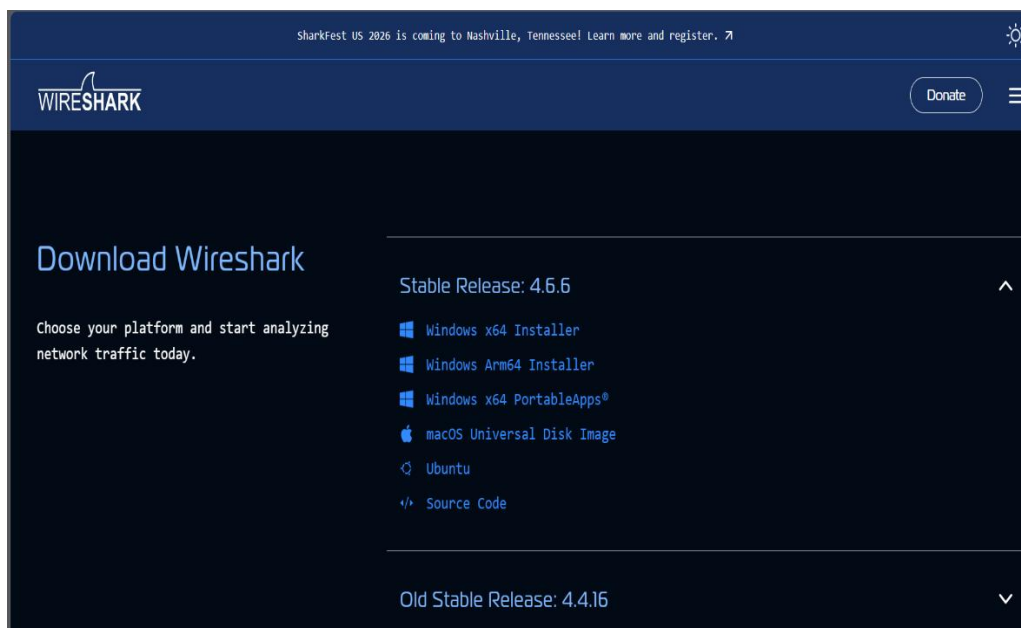
The installation included:

- Wireshark Network Analyzer
- Npcap Packet Capture Driver
- Protocol Support Components

The installation process completed successfully without errors.

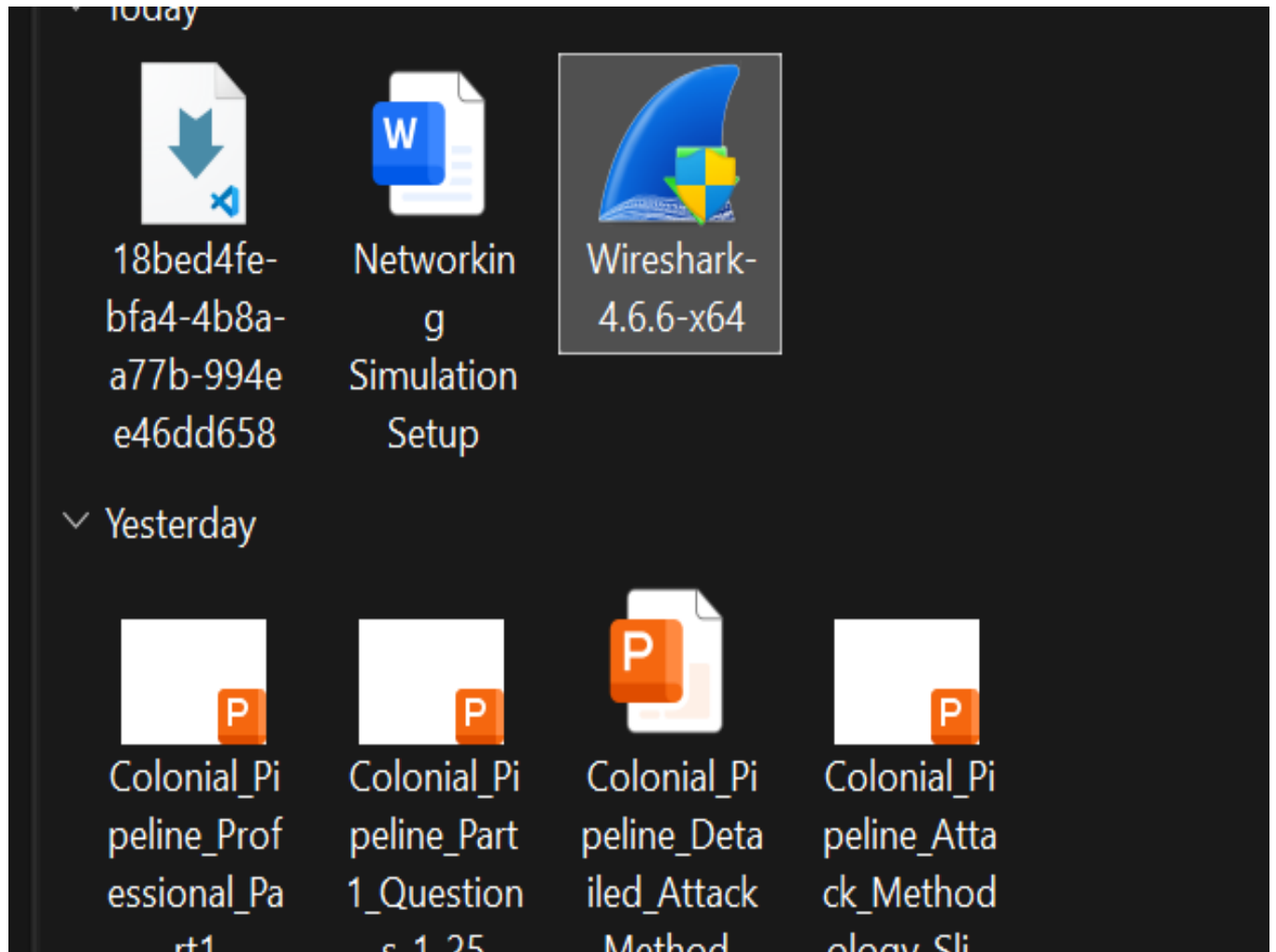
Screenshot 1

Wireshark Download Page



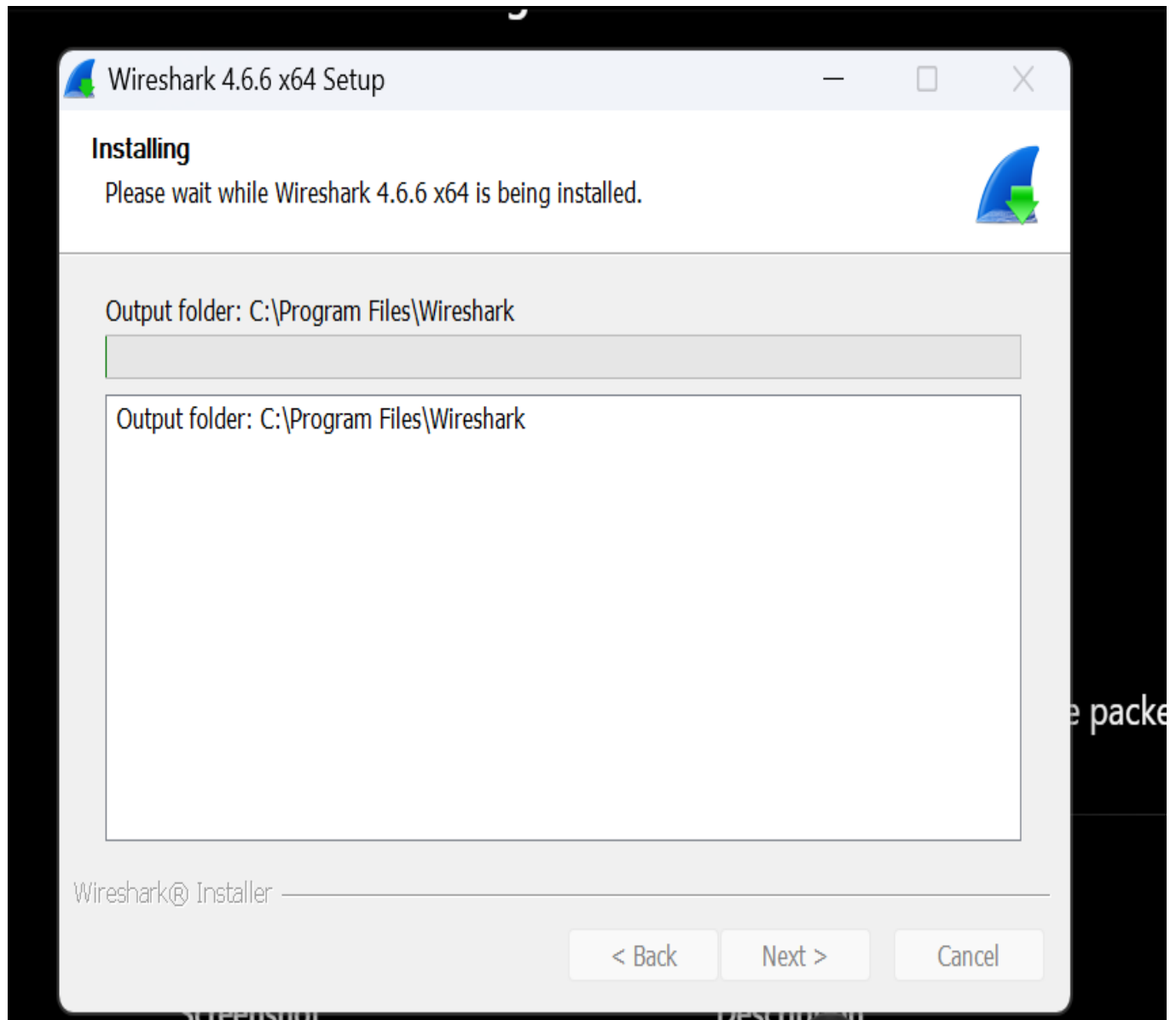
Screenshot 2

Downloaded Wireshark Installer



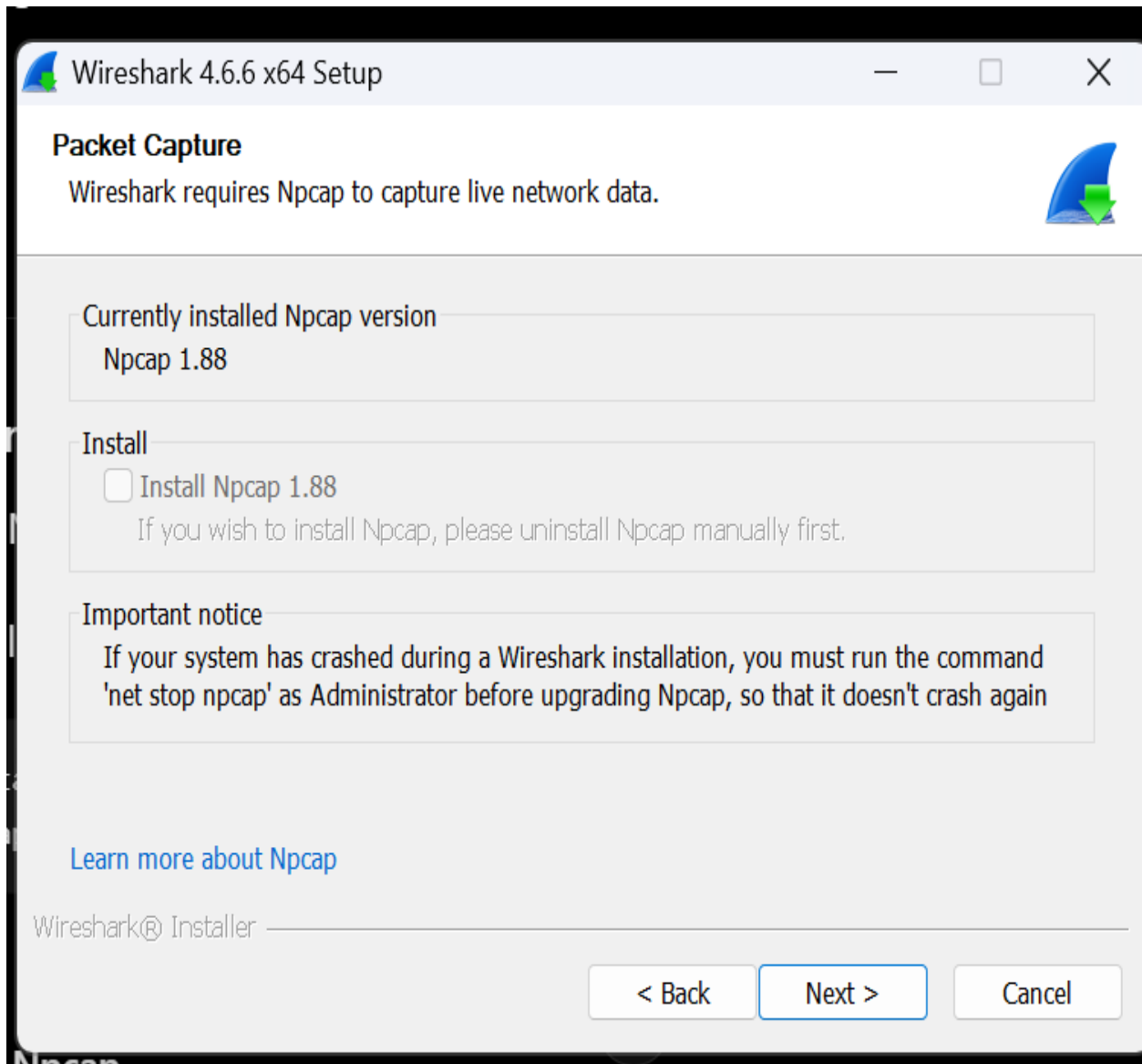
Screenshot 3

Wireshark Installation Wizard



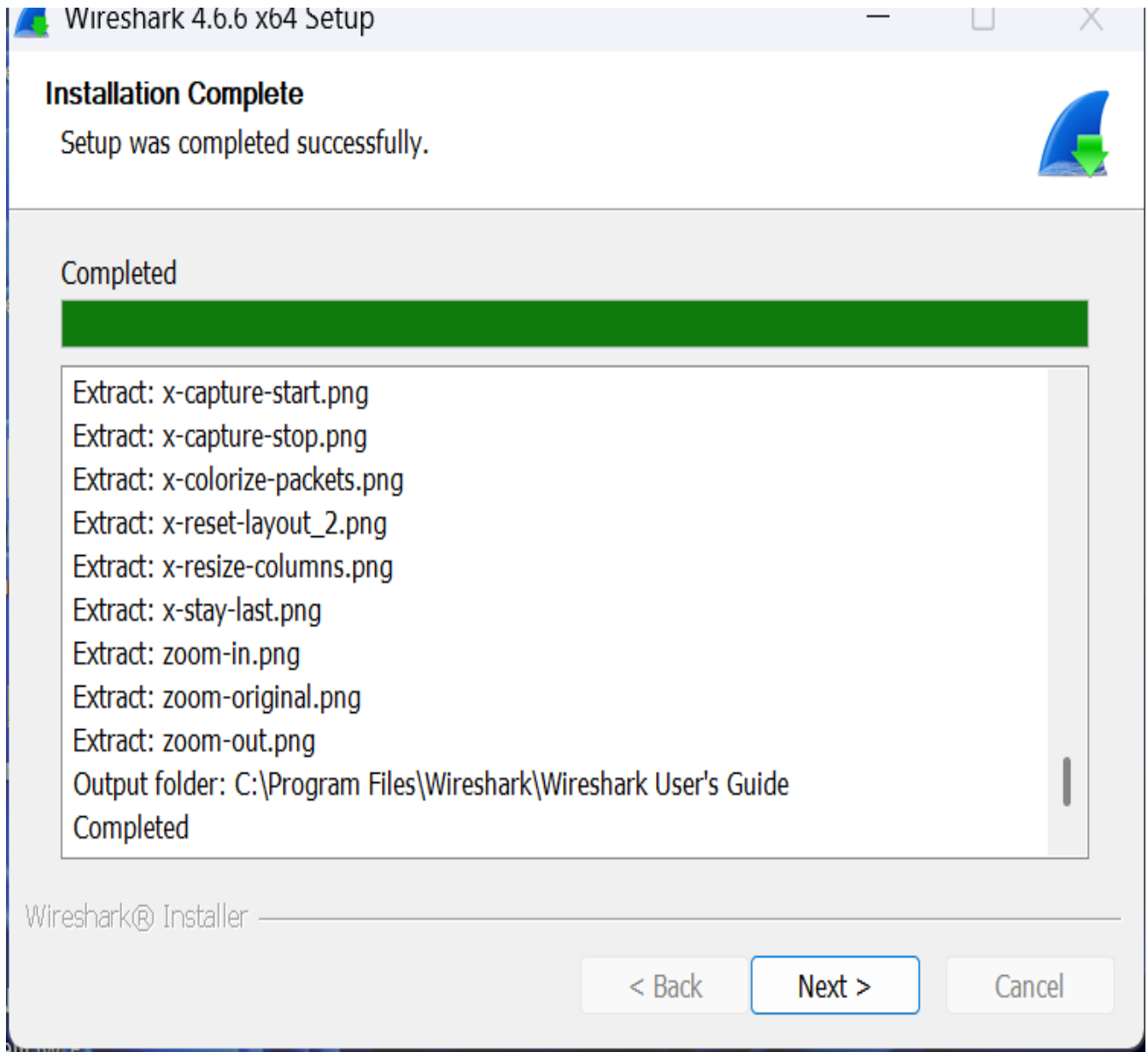
Screenshot 4

Npcap Configuration



Screenshot 5

Installation Completed Successfully



8. Network Interface Detection

After launching Wireshark, several network interfaces were detected.

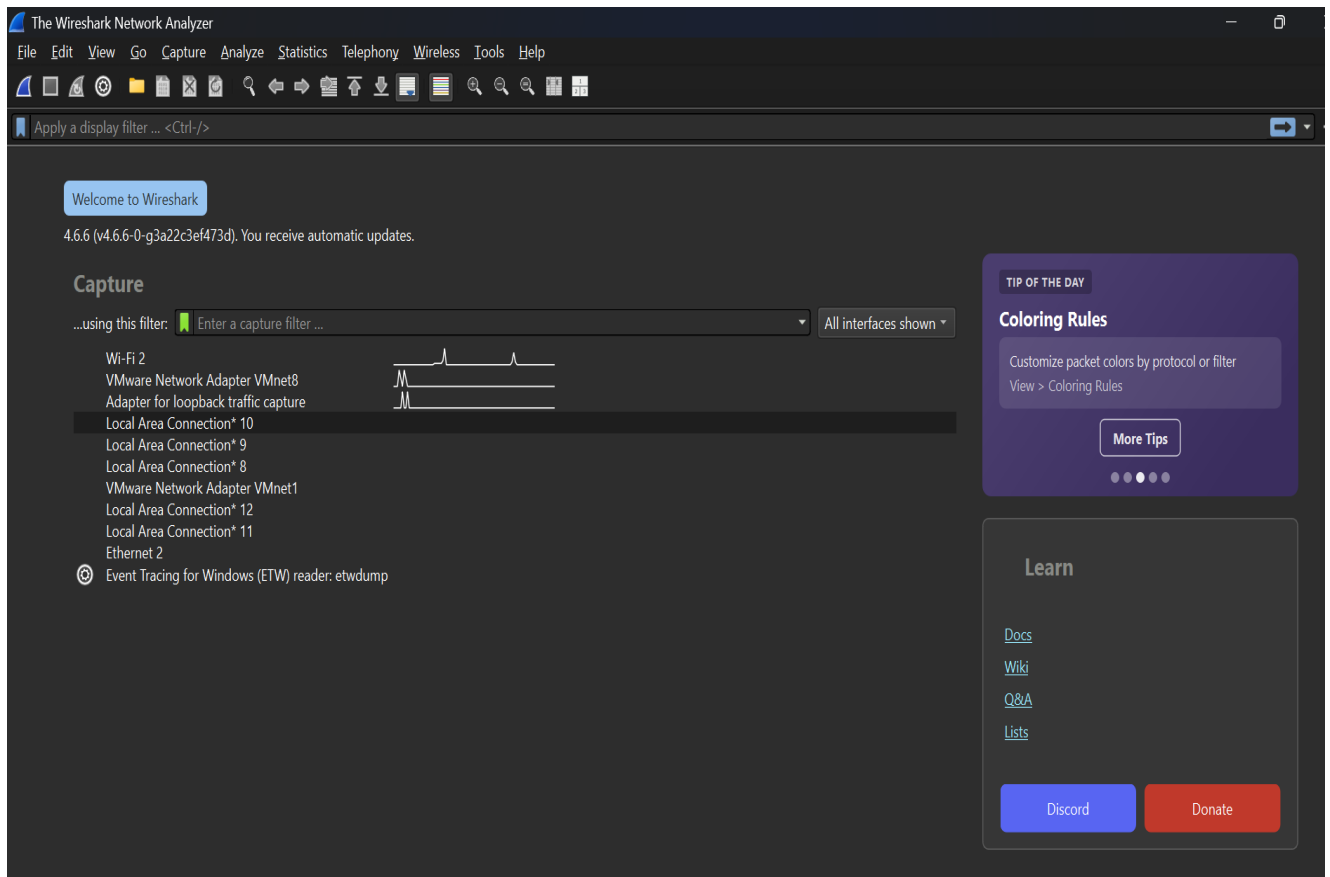
The primary active interface used for packet capture was:

Wi-Fi 2

This interface showed active traffic and was selected for monitoring.

Screenshot 6

Available Network Interfaces in Wireshark



9. Live Packet Capture

A live packet capture session was initiated on the Wi-Fi 2 network adapter.

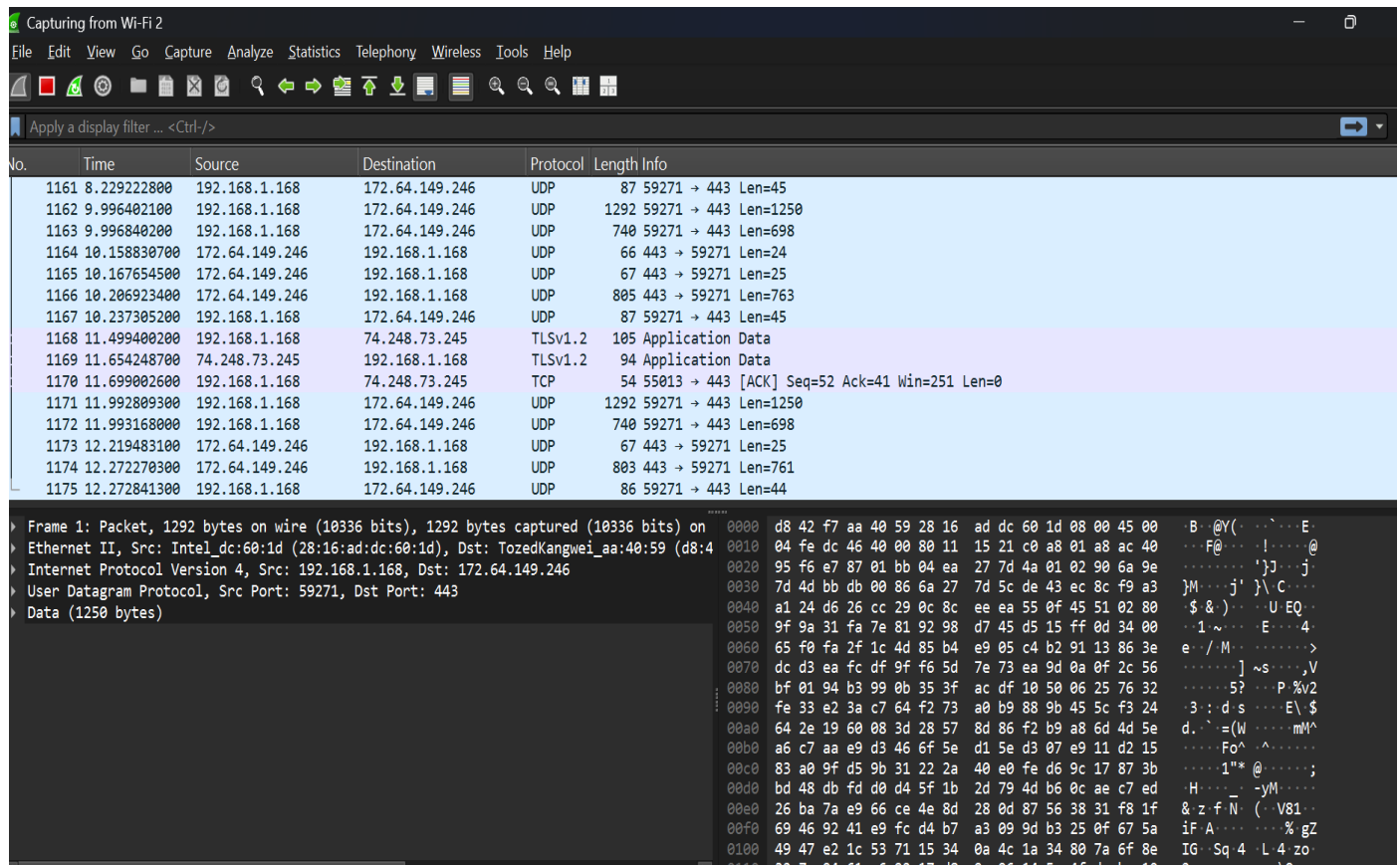
Traffic was generated through normal internet activities such as:

- Web browsing
- DNS lookups
- Background network communications

Wireshark successfully captured and displayed packets in real time.

Screenshot 7

Live Packet Capture



The screenshot displays the Wireshark interface with a live packet capture session. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for packet capture and analysis. A display filter bar shows "Apply a display filter ... <Ctrl-/>". The packet list pane shows a table of captured packets with columns for No., Time, Source, Destination, Protocol, and Length. The packet details pane shows the structure of the selected packet (Frame 1), including Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and Data. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1161	8.229222800	192.168.1.168	172.64.149.246	UDP	87	59271 → 443 Len=45
1162	9.996402100	192.168.1.168	172.64.149.246	UDP	1292	59271 → 443 Len=1250
1163	9.996840200	192.168.1.168	172.64.149.246	UDP	740	59271 → 443 Len=698
1164	10.158830700	172.64.149.246	192.168.1.168	UDP	66	443 → 59271 Len=24
1165	10.167654500	172.64.149.246	192.168.1.168	UDP	67	443 → 59271 Len=25
1166	10.206923400	172.64.149.246	192.168.1.168	UDP	805	443 → 59271 Len=763
1167	10.237305200	192.168.1.168	172.64.149.246	UDP	87	59271 → 443 Len=45
1168	11.499400200	192.168.1.168	74.248.73.245	TLSv1.2	105	Application Data
1169	11.654248700	74.248.73.245	192.168.1.168	TLSv1.2	94	Application Data
1170	11.699002600	192.168.1.168	74.248.73.245	TCP	54	55013 → 443 [ACK] Seq=52 Ack=41 Win=251 Len=0
1171	11.992809300	192.168.1.168	172.64.149.246	UDP	1292	59271 → 443 Len=1250
1172	11.993168000	192.168.1.168	172.64.149.246	UDP	740	59271 → 443 Len=698
1173	12.219483100	172.64.149.246	192.168.1.168	UDP	67	443 → 59271 Len=25
1174	12.272270300	172.64.149.246	192.168.1.168	UDP	803	443 → 59271 Len=761
1175	12.272841300	192.168.1.168	172.64.149.246	UDP	86	59271 → 443 Len=44

Frame 1: Packet, 1292 bytes on wire (10336 bits), 1292 bytes captured (10336 bits) on
Ethernet II, Src: Intel_dc:60:1d (28:16:ad:dc:60:1d), Dst: TozedKangwei_aa:40:59 (d8:4
Internet Protocol Version 4, Src: 192.168.1.168, Dst: 172.64.149.246
User Datagram Protocol, Src Port: 59271, Dst Port: 443
Data (1250 bytes)

0000 d8 42 f7 aa 40 59 28 16 ad dc 60 1d 08 00 45 00 :B @Y(.E
0010 04 fe dc 46 40 00 80 11 15 21 c0 a8 01 a8 ac 40 :..F@.. .!....@
0020 95 f6 e7 87 01 bb 04 ea 27 7d 4a 01 02 90 6a 9e :... ') . . . j
0030 7d 4d bb db 00 86 6a 27 7d 5c de 43 ec 8c f9 a3 :}M . . . j' } \ C . . .
0040 a1 24 d6 26 cc 29 0c 8c ee ea 55 0f 45 51 02 80 :\$ &) . . . U EQ . .
0050 9f 9a 31 fa 7e 81 92 98 d7 45 d5 15 ff 0d 34 00 :_ 1 ~ . . . E . . . 4
0060 65 f0 fa 2f 1c 4d 85 b4 e9 05 c4 b2 91 13 86 3e :e / - / M>
0070 dc d3 ea fc df 9f f6 5d 7e 73 ea 9d 0a 0f 2c 56 :.] ~ sV
0080 bf 01 94 b3 99 0b 35 3f ac df 10 50 06 25 76 32 :.5? . . . P %v2
0090 fe 33 e2 3a c7 64 f2 73 a0 b9 88 9b 45 5c f3 24 :3 : d sE \$
00a0 64 2e 19 60 08 3d 28 57 8d 86 f2 b9 a8 6d 4d 5e :d (WmM^
00b0 a6 c7 aa e9 d3 46 6f 5e d1 5e d3 07 e9 11 d2 15 :.Fo ^
00c0 83 a0 9f d5 9b 31 22 2a 40 e0 fe d6 9c 17 87 3b :.1"* @ ;
00d0 bd 48 db fd d0 d4 5f 1b 2d 79 4d b6 0c ae c7 ed :H-yM
00e0 26 ba 7a e9 66 ce 4e 8d 28 0d 87 56 38 31 f8 1f :& z f N (. . . . V81 . .
00f0 69 46 92 41 e9 fc d4 b7 a3 09 9d b3 25 0f 67 5a :iF A% gZ
0100 49 47 e2 1c 53 71 15 34 0a 4c 1a 34 80 7a 6f 8e :IG Sq 4 . L 4 zo
0110 32 7a 94 61 c6 82 17 d8 9a 86 14 5c 4f dc ba 10 :7 a10

10. Protocol Analysis

During the capture process, several protocols were observed.

10.1 DNS Analysis

Protocol

DNS (Domain Name System)

Purpose

DNS converts domain names into IP addresses.

Example:

google.com



74.125.128.84

Observed DNS traffic included:

- DNS Queries
- DNS Responses
- Hostname Resolution

Screenshot 8

DNS Traffic Analysis

No.	Source	Destination	Protocol	Length	Info
15355	192.168.1.1	192.168.1.168	DNS	136	Standard query response 0xf058 HTTPS waa-pa.clients6.google.com SOA ns1.google.com
15273	653.338962100	192.168.1.1	DNS	125	Standard query response 0xd448 HTTPS play.google.com SOA ns1.google.com
15274	653.338962700	192.168.1.1	DNS	91	Standard query response 0xc22e A play.google.com A 172.217.20.46
15275	653.338963200	192.168.1.1	DNS	102	Standard query response 0x16d6 A waa-pa.clients6.google.com A 142.251.209.106
15352	654.250489800	192.168.1.1	DNS	79	Standard query 0x24c4 HTTPS accounts.google.com
15353	654.250933100	192.168.1.1	DNS	79	Standard query 0xfafa A accounts.google.com
15354	654.252381800	192.168.1.1	DNS	129	Standard query response 0x24c4 HTTPS accounts.google.com SOA ns1.google.com
15355	654.252381800	192.168.1.1	DNS	92	Standard query response 0xfafa A accounts.google.com A 74.125.128.84
15679	744.020427800	192.168.1.1	DNS	76	Standard query 0xc452 HTTPS web.whatsapp.com
15680	744.020744700	192.168.1.1	DNS	76	Standard query 0xa9f2 A web.whatsapp.com
15681	744.067803200	192.168.1.1	DNS	129	Standard query response 0xa9f2 A web.whatsapp.com CNAME mmx-ds.cdn.whatsapp.net A 1...
15682	744.071349200	192.168.1.1	DNS	158	Standard query response 0xc452 HTTPS web.whatsapp.com CNAME mmx-ds.cdn.whatsapp.net...
15742	748.515210600	192.168.1.1	DNS	90	Standard query 0x958f HTTPS m365cdn.nel.measure.office.net
15743	748.516335500	192.168.1.1	DNS	90	Standard query 0x029b A m365cdn.nel.measure.office.net
15744	748.547741500	192.168.1.1	DNS	233	Standard query response 0x958f HTTPS m365cdn.nel.measure.office.net CNAME nel.measu...

Frame 15355: Packet, 95 bytes on wire (769 bits), 95 bytes captured (769 bits) on interface 0
Ethernet II, Src: Intel_Dc:60:1d:28:11, Dst: Intel_Dc:60:1d:28:11
Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.168
User Datagram Protocol, Src Port: 53, Dst Port: 61328
Domain Name System (response)

0000 28 16 ad dc 68 1d d8 12 f7 aa 4b 53 08 00 45 00 (.....B...E...)
0010 00 51 34 90 40 00 00 11 82 dc c0 a8 01 01 c0 a8 Q4 @ @
0020 01 a8 00 35 ef 90 00 3d 46 aa fa 81 80 00 015...F.....
0030 00 01 00 00 00 00 08 61 63 63 6f 75 6e 74 73 06a counts...
0040 67 6f 6f 67 6c 65 03 63 6f 6d 00 00 01 00 01 c0 google c om.....
0050 0c 00 01 00 01 00 00 00 6f 00 04 4a 7d 80 54o J) T.....

Security Importance

DNS monitoring helps identify:

- Malicious domains
- Command and Control Servers
- DNS Tunneling

10.2 ICMP Analysis

Protocol

ICMP (Internet Control Message Protocol)

Purpose

Used for:

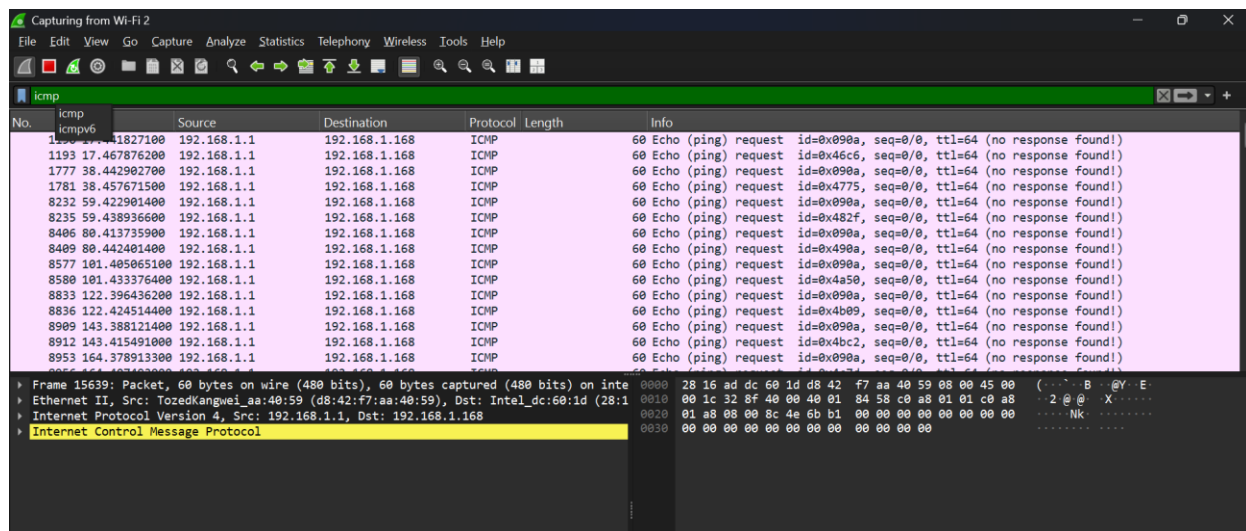
- Ping Requests
- Ping Replies
- Network Diagnostics

Observed traffic included:

- Echo Requests

Screenshot 9

ICMP Traffic Analysis



Security Importance

ICMP is useful for:

- Network troubleshooting
- Host discovery
- Availability testing

Attackers may also use ICMP for reconnaissance.

10.3 TCP Analysis

Protocol

TCP (Transmission Control Protocol)

Purpose

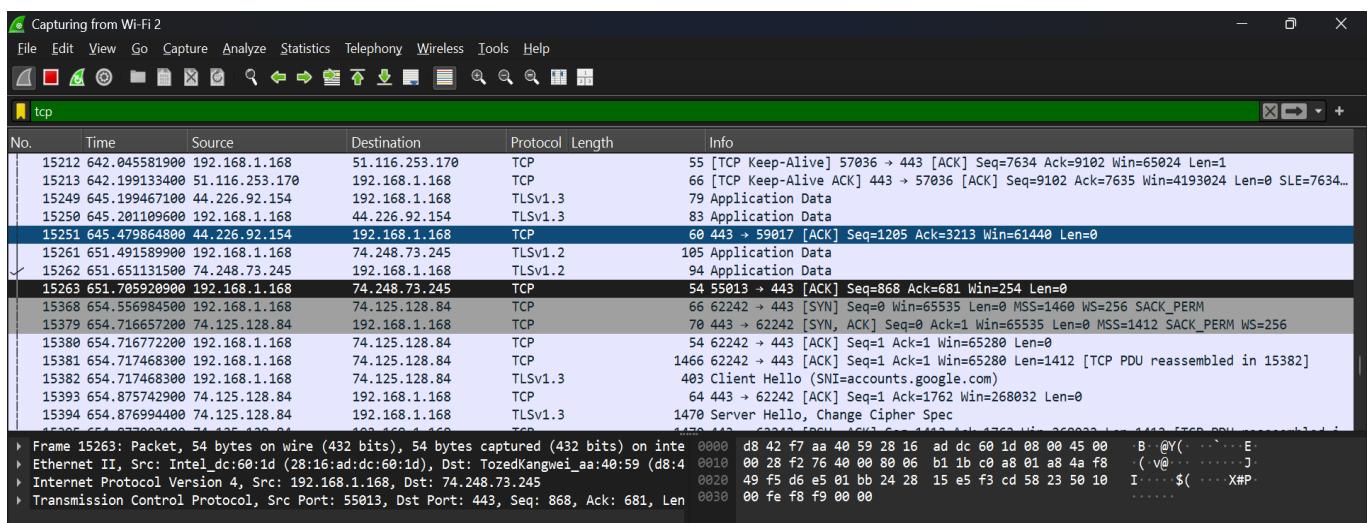
Provides reliable communication between devices.

Observed TCP activities included:

- TCP Handshake
- ACK Packets
- Secure Communications

Screenshot 10

TCP Traffic Analysis



The screenshot shows a Wireshark packet capture window titled "Capturing from Wi-Fi 2". The packet list on the left shows several packets, with packet 15251 selected. The packet details pane on the right shows the structure of the selected packet, which is a TCP segment. The packet is from source 192.168.1.168 to destination 192.168.1.168, port 443. The details pane shows the TCP header and application data. The packet bytes pane at the bottom shows the raw data of the packet, including the Ethernet II header, IP header, and TCP segment.

No.	Time	Source	Destination	Protocol	Length	Info
15212	642.045581900	192.168.1.168	51.116.253.170	TCP	55	[TCP Keep-Alive] 57036 → 443 [ACK] Seq=7634 Ack=9102 Win=65024 Len=1
15213	642.199133400	51.116.253.170	192.168.1.168	TCP	66	[TCP Keep-Alive ACK] 443 → 57036 [ACK] Seq=9102 Ack=7635 Win=4193024 Len=0 SLE=7634...
15249	645.199467100	44.226.92.154	192.168.1.168	TLSv1.3	79	Application Data
15250	645.201109600	192.168.1.168	44.226.92.154	TLSv1.3	83	Application Data
15251	645.479864800	44.226.92.154	192.168.1.168	TCP	60	443 → 59017 [ACK] Seq=1205 Ack=3213 Win=61440 Len=0
15261	651.491589900	192.168.1.168	74.248.73.245	TLSv1.2	105	Application Data
15262	651.651131500	74.248.73.245	192.168.1.168	TLSv1.2	94	Application Data
15263	651.705920900	192.168.1.168	74.248.73.245	TCP	54	55013 → 443 [ACK] Seq=868 Ack=681 Win=254 Len=0
15368	654.556984500	192.168.1.168	74.125.128.84	TCP	66	62242 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
15379	654.716657200	74.125.128.84	192.168.1.168	TCP	70	443 → 62242 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
15380	654.716772200	192.168.1.168	74.125.128.84	TCP	54	62242 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
15381	654.717468300	192.168.1.168	74.125.128.84	TCP	1466	62242 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1412 [TCP PDU reassembled in 15382]
15382	654.717468300	192.168.1.168	74.125.128.84	TLSv1.3	403	Client Hello (SNI=accounts.google.com)
15393	654.875742900	74.125.128.84	192.168.1.168	TCP	64	443 → 62242 [ACK] Seq=1 Ack=1762 Win=268032 Len=0
15394	654.876994400	74.125.128.84	192.168.1.168	TLSv1.3	1470	Server Hello, Change Cipher Spec

Frame 15263: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface 0
Ethernet II, Src: Intel_dc:60:1d (28:16:ad:dc:60:1d), Dst: TozedKangwei_aa:40:59 (d8:40:59:00:28:f2:76)
Internet Protocol Version 4, Src: 192.168.1.168, Dst: 74.248.73.245
Transmission Control Protocol, Src Port: 55013, Dst Port: 443, Seq: 868, Ack: 681, Len: 0

Security Importance

TCP analysis helps identify:

- Connection establishment
- Network anomalies
- Suspicious communications

11. Security Monitoring Findings

The following protocols were identified during analysis:

Protocol	Purpose
TCP	Reliable communication
DNS	Name resolution
ICMP	Connectivity testing
TLS	Secure encrypted communication
UDP	Fast communication transport

The captured traffic demonstrated normal network behavior.

No suspicious traffic was observed during the capture session.

12. Testing and Validation

The following tests were successfully completed:

Test	Result
Wireshark Installation	PASS
Npcap Installation	PASS
Interface Detection	PASS
Live Packet Capture	PASS
DNS Analysis	PASS
ICMP Analysis	PASS
TCP Analysis	PASS

13. Cybersecurity Relevance

Wireshark is widely used by:

- Security Analysts
- SOC Analysts
- Network Engineers
- Penetration Testers
- Incident Response Teams

The tool assists in:

- Threat Hunting
- Malware Investigation
- Traffic Analysis
- Intrusion Detection
- Network Troubleshooting

14. Conclusion

The project successfully established a network monitoring environment using Wireshark. Live network traffic was captured and analyzed through the Wi-Fi network interface. Multiple protocols including DNS, ICMP, TCP, UDP, and TLS were identified and examined.

The project provided practical experience in packet capture, protocol analysis, and cybersecurity monitoring. These skills are essential for cybersecurity professionals involved in network security, incident response, digital forensics, and security operations.